

Set 9: Atomic structure and the periodic table

Notes

An electron configuration describes the arrangement of electrons surrounding the nucleus of an atom. Electrons occupy regions of space around the nucleus which we refer to as shells. Electrons in different shells have different energies. The energy of the electrons depends upon their distance from the nucleus and the shell they occupy. The electron configuration indicates how many electrons are in each energy level or shell.

Electrons, in their ground state, occupy the lowest energy levels or shells possible. Further away from the nucleus, the energy levels start to overlap to some degree and so some electrons may occupy a higher energy level before the lower energy level is full. Noble gases have stable electron configurations.

Examples

1. Oxygen: atomic number (Z) = 8, there will also be eight electrons in a neutral atom of oxygen, two in the first energy level and the remaining six in the second energy level. The electron configuration of oxygen would be written as: 2, 6
2. Oxide ion: O^{2-} , $Z = 8$, there will be eight electrons plus two extra electrons as it has a negative two charge. Therefore there will be two electrons in the first energy level and the remaining eight will be in the second energy level. The electron configuration of the oxide ion would be written as: 2, 8

Set 9: Exercises

1. Copy and complete a table similar to the following to show details of the first 20 elements. As an example the element nitrogen has been done for you.

Z	Name	Symbol	Metal/non-metal	Electron configuration	Valence electron behaviour
1					
2					
3					
4					
5					
6					
7	Nitrogen	N	Non-metal	2, 5	Shares e^- or gains 3 e^-
8					

2. (a) Which elements appear to always lose electrons?
(b) Describe the position of these elements on the Periodic Table.
3. (a) Which elements appear to always gain electrons?
(b) Describe the position of these elements on the Periodic Table.
4. (a) Which elements do not gain or lose electrons?
(b) Describe the position of these elements on the Periodic Table.
5. Is there a pattern between the number of valence electrons and whether or not electrons are gained or lost?

6. Complete the following table:

	Symbol	Atomic number	Number of protons	Number of electrons	Number of neutrons
Example	${}^{19}_9\text{F}$	9	9	9	10
A	${}^{16}_8\text{X}$				
B	${}^{17}_8\text{X}$				
C	${}^{16}_8\text{X}^{2-}$				
D	${}^{17}_9\text{X}$				

7. In the table in question 6, which species are isotopes of each other? Explain your answer.
8. There is a saying that protons give an atom its identity and electrons give an atom its personality. Explain what this means.
9. Write the electron configurations for the following species:
- S^{2-}
 - Al^{3+}
 - K
 - C
10. Which elements are represented by the following electron configurations?
- 2, 8, 7
 - 2, 8, 8, 2
 - 2, 8, 3
 - 2, 8, 8, 8
11. Which of the following electron configurations represent elements in which electrons are not in the ground state?
- 2, 7, 8, 1
 - 1, 8, 8
 - 2, 8, 7, 1
 - 2, 8, 8, 1, 5